Combining IQ Scores

Analysis and a corrected statistical treatment

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The covariance identities reproduced in Ronald K. Hoeflin's article are correct. Their application to qualifying IQ scores requires additional assumptions about the reference population, the composite's distribution, and the meaning of the score. This report distinguishes those assumptions from the algebra, corrects the numerical table, and supplies an independently written replacement article. All displayed numerical examples have been recalculated. A technical appendix illustrates why a standardized composite and an estimate of latent ability need not coincide.

1 Analysis of the original article

Hoeflin reproduces George A. Ferguson's treatment of variances and covariances and applies it to combinations of IQ scores. The basic identity

$$\operatorname{Var}(X+Y) = \operatorname{Var}(X) + \operatorname{Var}(Y) + 2\operatorname{Cov}(X,Y) \tag{1}$$

is correct. It determines the variance of a sum without requiring normality. The principal weaknesses arise when a standardized sum is assigned an extreme population percentile and used as a qualifying score.

- **Population calibration.** Correlations among the article's 27 selected high scorers do not establish the correlations in the population used to standardize the tests. Selection can change the relationships, and the appropriate correction depends on how selection occurred (Sackett and Yang, 2000).
- **Distributional assumptions.** Mean zero and variance one do not establish a normal distribution. A one-in-a-million upper-tail claim requires additional justification for the tail model.
- **Numerical correction.** The 100-test entry should be about 3.69, rather than 3.71, standard deviations. The more accurate normal quantile also changes some final digits elsewhere; for example, the four-test result rounds to 3.98 rather than 3.97.
- **Score interpretation.** A standardized composite, a latent ability estimate, and a validated IQ score are distinct quantities. Their equivalence requires evidence about the battery and its intended use (American Educational Research Association et al., 2014, Standard 1.14).
- **Threshold precision.** Rounded whole-number scores are useful illustrations, but are not exact qualifying cutoffs. The scale's standard deviation and the classification rule must be specified.

The replacement below preserves the valid calculation while making its assumptions explicit. The article is newly written; it is not a transcription or a reproduction of the original scans.

2 Corrected article: a statistical method and its limits

Several cognitive test scores can be combined into a single score. The calculation must account for the tests' correlations: correlated scores cannot be treated as independent when calculating the composite's variance. The resulting composite also needs an explicit interpretation. Rescaling it to an IQ-style scale supplies units; evidence about the battery establishes what those units mean.

Before calculating a composite, specify the tests, their weights, the administration procedure, and the population against which scores will be compared. Composite construction and interpretation require their own justification; evidence supporting the individual tests does not automatically establish the meaning of their combination (American Educational Research Association et al., 2014, Standard 1.14).

2.1 Put the component scores on comparable scales

For test i, convert the observed score x_i to a standardized score:

$$z_i = \frac{x_i - \mu_i}{\sigma_i}, \tag{2}$$

where μ_i and σ_i are its mean and standard deviation in the intended reference population. The norms must support comparison with that population. Using different numbers labelled "IQ" does not establish their comparability.

For example, a score of 160 on a scale with mean 100 and standard deviation 15 corresponds to $z = 4$. The same numerical score on a scale with standard deviation 16 corresponds to $z = 3.75$. These scores should be standardized before they are combined.

2.2 Standardize the weighted composite

Let R be the population correlation matrix of the standardized test scores, and let w contain fixed weights. The weighted sum has variance

$$\mathrm{Var}(w^\mathsf{T} Z) = w^\mathsf{T} R w. \tag{3}$$

Consequently, provided this variance is positive, a standardized composite is

$$\boxed{Z_C = \frac{w^\mathsf{T} z}{\sqrt{w^\mathsf{T} R w}}.} \tag{4}$$

With the correct population parameters, this composite has mean zero and variance one. This statement requires finite variances, but does not require normality. When the parameters are estimated, their uncertainty affects the calibration of the composite.

For an equally weighted battery of $n \geq 2$ tests, write $\bar{z}$ for the average standardized score and $\bar{r}$ for the average correlation across the distinct pairs of tests:

$$\bar{r} = \frac{2}{n(n-1)} \sum_{i<j} r_{ij}. \tag{5}$$

Then

$$\begin{aligned}\mathrm{Var}(\bar{Z}) &= \frac{n + 2\sum_{i<j} r_{ij}}{n^2} \\ &= \frac{1 + (n-1)\bar{r}}{n} = \bar{r} + \frac{1-\bar{r}}{n}.\end{aligned} \tag{6}$$

Therefore,

$$\boxed{Z_C = \frac{\bar{z}}{\sqrt{\bar{r} + (1-\bar{r})/n}}.} \tag{7}$$

For a single test, $Z_C = z_1$; no inter-test correlation is needed.

Identical pairwise correlations are unnecessary for equation (6); their average is sufficient for this particular equal-weight composite. The full matrix remains relevant to other weighting schemes and to understanding the battery.

An IQ-style reporting scale with mean 100 and standard deviation s can be defined by

$$I_C = 100 + sZ_C. \tag{8}$$

When all component scores already use mean 100 and the same standard deviation s, the formula becomes

$$I_C = 100 + \frac{\bar{I} - 100}{\sqrt{\bar{r} + (1-\bar{r})/n}}, \tag{9}$$

where $\bar{I}$ is their arithmetic average.

For example, three scores averaging 160, each on an SD-16 scale, produce $\bar{z} = 3.75$. If their population average correlation is 0.60, then

$$Z_C = \frac{3.75}{\sqrt{0.60 + 0.40/3}} \approx 4.3791, \qquad I_C \approx 170.06. \tag{10}$$

This is a score on the specified composite. Interpreting it as a measure of general intelligence requires evidence about what the composite measures.

2.3 Separate standardization from percentile estimation

To assign a percentile, we also need the composite's distribution. Mean zero and variance one do not make a distribution normal. Joint multivariate normality of the component scores is a sufficient condition for a fixed linear composite to be normal. Normal distributions for the individual tests alone are insufficient.

If a standard-normal distribution is justified for the composite, its percentile is $100\Phi(Z_C)$, where Φ is the standard-normal cumulative distribution function. A one-in-a-million upper-tail probability corresponds to

$$q = \Phi^{-1}(0.999999) \approx 4.7534243088. \tag{11}$$

Thus the model-based 99.9999th-percentile threshold is

$$\boxed{\bar{z}_{\min} = q\sqrt{\bar{r} + \frac{1-\bar{r}}{n}}.} \tag{12}$$

The quantile can be reproduced in R with `qnorm(1e-6, lower.tail = FALSE)`; the official documentation defines the normal quantile function and its upper-tail option (R Core Team).

For $\bar{r} = 0.60$, the thresholds are shown in table 1.

Table 1: Model-based 99.9999th-percentile thresholds, assuming a normal composite and average pairwise correlation 0.60.

Number of tests	**Required average component z**	**Required average component score, mean 100 and SD 16**
1	4.7534	176.05
2	4.2516	168.03
3	4.0706	165.13
4	3.9770	163.63
5	3.9198	162.72
10	3.8027	160.84
100	3.6942	159.11
1,000	3.6832	158.93
Limit as n increases	3.6820	158.91

These are thresholds for the *average component score*, under the stated model. The target composite score remains approximately 176.05 on the SD-16 scale in every row. Values are rounded for display; an operational classification rule must specify rounding and use the appropriate attainable scores.

For an SD-15 reporting scale, use $100 + 15\bar{z}_{\min}$ instead. In particular, the single-test threshold would be approximately 171.30. An IQ number without its scale convention is ambiguous.

2.4 Interpret the limiting threshold and correlation sensitivity

The limiting average threshold follows from

$$\lim_{n\to\infty} \bar{z}_{\min} = q\sqrt{\bar{r}}, \tag{13}$$

if the average correlation remains fixed as tests are added. With $\bar{r} = 0.60$, the limiting average component score on the SD-16 scale is approximately 158.91. This is a feature of the assumed sequence of batteries and its threshold; it is not a ceiling on possible composite scores. Adding tests can also change the average correlation, so the fixed-correlation limit should not be treated as a universal rule.

The choice of correlation matters substantially. For three tests, an average correlation of 0.90 raises the required average SD-16 component score to approximately 173.48, compared with 165.13 at 0.60. A smaller assumed correlation gives a larger composite score for the same positive average performance. The correlation therefore needs to be estimated for the relevant population and administration procedure. A lower correlation does not, by itself, establish better measurement; it can also reflect different constructs or greater measurement noise.

2.5 Calibrate the population and the scoring procedure

Correlations in a selected high-scoring sample need not equal correlations in the population used to standardize the tests. Selection can change score variances and relationships. The appropriate treatment depends on the selection mechanism and the information available about the unrestricted population; there

is no universal adjustment. Sackett and Yang (2000) explain why range-restriction corrections depend on those details. A small selected sample cannot directly establish a one-in-a-million population frequency.

The scoring procedure also matters. A fixed battery, the best few results from many attempts, and a rule that keeps adding tests until someone qualifies have different score distributions. Each procedure requires calibration for the rule actually used. Repeated administrations can introduce practice effects, as documented by Hausknecht et al. (2007). Repeated or overlapping tests should not be counted as fresh independent observations without accounting for their dependence and administration history.

Finally, variability across people is different from uncertainty about one person. The correlation matrix determines how the composite varies in the reference population. It does not, by itself, provide an individual confidence interval, separate general ability from shared methods, or establish measurement precision at extreme scores. Those questions require an appropriate measurement model and supporting data.

A defensible report should identify the battery, norms, weights, estimated correlations, and uncertainty. Direct composite norms should be used when they adequately support the intended interpretation. If an extreme percentile comes from a distributional model beyond the available normative evidence, report it as a model-based extrapolation and show how sensitive it is to the assumptions. The testing standards require clear descriptions of reference populations and norming procedures in Standards 5.8–5.9, and justification for score interpretations in Standards 1.1–1.4 (American Educational Research Association et al., 2014).

A Composite scores and latent ability: a mathematical illustration

To see why these quantities differ, suppose, expressly as a hypothetical model, that

$$Z_i = \sqrt{\rho}\, G + \sqrt{1-\rho}\, \varepsilon_i, \qquad 0 < \rho < 1, \tag{14}$$

where G and all ε_i are independent standard-normal variables. This model treats all shared variance as one common factor and all residuals as mutually independent. Real batteries need not satisfy those assumptions.

In this particular model, the equal-weight composite's squared correlation with G is

$$\mathcal{R} = \mathrm{Corr}(G, Z_C)^2 = \frac{n\rho}{1 + (n-1)\rho}. \tag{15}$$

The conditional distribution of the latent factor has

$$\mathrm{E}[G \mid Z_1, \ldots, Z_n] = \sqrt{\mathcal{R}}\, Z_C, \tag{16}$$

$$\mathrm{Var}(G \mid Z_1, \ldots, Z_n) = \frac{1-\rho}{1 + (n-1)\rho}. \tag{17}$$

These identities follow from Gaussian conditioning. In particular, $\mathrm{Cov}(G, \bar{Z}) = \sqrt{\rho}$ and $\mathrm{Var}(\bar{Z}) = \rho + (1-\rho)/n$; the residual contrasts contain no additional information about G under this model.

For three tests with $\rho = 0.60$, a composite at $Z_C = 4.7534$ has conditional mean G of approximately 4.30 and conditional standard deviation approximately 0.43. On a hypothetical latent scale with mean 100 and SD 16, these become a mean of approximately 168.79 and a standard deviation of 6.82. The composite's reporting-scale value is approximately 176.05. The two values answer different questions, even in this unusually simple model.

The illustration is a mathematical counterexample to treating a standardized composite as automatically identical to a latent-ability estimate. It is not a proposed correction factor for actual IQ scores, and inter-test correlation should not generally be equated with reliability or common-factor variance.

B Numerical reproduction

The accompanying script `scripts/verify_calculations.py` recalculates the normal quantile, checks the table against `data/thresholds.csv` and the numerical entries in this source, and verifies the worked examples. It uses the upper-tail probability directly through `scipy.stats.norm.isf(1e-6)`. The mathematical results do not depend on the choice of software.

C Expanded software verification and its limits

The original verification selected only the arithmetic core. That boundary was broader than necessary: the calculation and simulation drivers and most numerical orchestration can be analyzed in SPARK. The expanded implementation makes `Hoeflin_Calculations`, `Hoeflin_Simulation`, and the algorithm bodies in `Normal_Support` SPARK-on, alongside the unchanged `Hoeflin_Core`. Argument parsing and CSV rendering are isolated in `Report_IO`, whose SPARK interface explicitly models external state. The proof project enumerates all eight source files, including the trusted I/O completion, rather than selecting only the core.

The expanded computer10 evidence report records the current source hashes, exact commands, tool identities, native reports, coverage, numerical comparisons, and Python checks. This campaign was completed on 1 November 2026; the directory name retains the date on which the expansion began. Earlier 47-check and 110-test results remain immutable historical snapshots, not counts for the expanded implementation.

C.1 What is now analyzed

SPARK now covers the inverse-survival bisection loop, exact 64-bit random state recurrence, open-unit conversion, Box–Muller composition, common-factor simulation, signed and squared accumulations, event-count updates, and both main procedures. Random generation uses procedures with explicit state updates instead of side-effecting functions. Proved loop invariants and small SPARK helpers bound accumulation; the exact integer-to-floating successor lemma is proved, not imported as an axiom. The original arithmetic core still establishes exact integer variance bookkeeping, bounded floating-point transforms, and nested event counts with specified increments.

The fresh five-prover portfolio discharged **209 native checks: 43 flow checks and 166 prover checks**, with zero unproved and zero justified checks. CVC5, Z3, Alt-Ergo, classic COLIBRI, and Yices were requested together. This result does not imply that every solver independently completed every obligation. The deliberately incorrect inclusion-exclusion control was rejected with an unproved functional contract. Fresh source copies and object directories prevent reuse of stale proof results. The compiler identifies itself as GNAT 13.2.0; the matching GNATprove banner is `0.0w`. Exact executable identities are retained in the manifest.

Native coverage records 22 `all`, 14 `spec`, and zero `no` entities. Here `all` describes a SPARK specification and body, or a declaration requiring no body; it is not itself a proof result. The `spec` entries expose trusted completions. In the earlier source, whole off-mode units emitted empty entity arrays, so zero `no` entries must never be interpreted as zero excluded code. The retained flow-only baseline has nine `all` entities and is explicitly not a new proof campaign. Native proof totals and coverage statuses are audited separately.

C.2 Trusted numerical and I/O boundaries

Only five small numerical completions in `Normal_Support` remain off: `Square_Root`, `Survival`, `Check_Inverse_Bracket`, `Rare_Quantile`, and `Checked_Radius`. Their bodies check each promised normal-return range or relationship and raise on rejection; they do not clamp values. The foreign C `erfc` import and `Report_IO`'s parsing, formatting, output, and exceptions remain trusted. The installed Ada elementary-function specifications are SPARK-compatible, but their limited contracts and unproved implementations remain part of the proof boundary.

Consequently, a caller proof is conditional on these interfaces: it does not prove numerical backend accuracy, that a guard cannot reject, or that I/O succeeds. Bisection bounds do not prove survival-function monotonicity, inverse convergence, or an approximation-error bound. No project `pragma Assume`, imported proof axiom, or suppressed check is used. Zero explicit assumptions or suppressions in native metadata does not eliminate trust in off-mode completions or library contracts.

C.3 Numerical regression, Deal, and audit results

Fresh Ada calculation and simulation outputs match both retained CSVs byte for byte. The simulation uses 1,000,000 trials for each of eight test-count cases with seed `20261030` reset between cases. Independent SciPy validation and Python replay pass normally and under `python -O`; tail counts match exactly, and means and variances satisfy the recorded tolerances. Minimum valid trial and boundary-seed runs also succeed, while invalid trial counts are rejected. Numerical equivalence of the core's proof-friendly evaluation orders is checked against the report formulas; bit-identical floating results are claimed only for the explicitly compared retained CSVs.

Every current Python file and named function, including tests and evidence tools, carries Deal contracts. Repository-wide coverage, clean positive static lint, the expected `DEL012` negative control, runtime/property tests, numerical scripts, and explicit historical-source evidence audits pass in normal and optimized Python. The final inventory has 22 files, 177 named functions, and 502 Deal decorators with zero gaps; all 21 positive files have zero lint diagnostics, and 164 tests pass in each execution mode. Deal Solver proves the two pure acceptance-gate postconditions and rejects the incorrect symbolic control. These focused symbolic results do not prove the full Python programs; clean lint and finite tests are also not universal proofs. CLI entry points explicitly enable Deal under optimized Python.

The current evidence auditor checks both required projects, required SPARK-on algorithm/driver entities, native totals, raw coverage, source hashes, checksums, and negative controls. Historical Ada campaigns require an explicitly selected matching source tree; there is no automatic Ada fallback. Historical Python may be resolved through verified immutable Git blobs. An evidence audit checks record integrity and source identity; it does not rerun the provers or independently validate SMT proof objects.

Common random numbers make the eight simulation cases dependent. Replay does not establish independent draws; rare counts at this sample size do not validate a one-in-a-million tail. Neither Deal nor SPARK proves the Gaussian common-factor model, empirical covariance calibration, an identification of correlation with reliability, or a psychometric interpretation. The distinction between a standardized composite and latent ability remains essential.

References

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